



CSE 320 - Computer Networks

Example MT Questions

Q1. Fill in the Space

1. The Internet is often described as a **network of** _____ that interconnects millions of hosts.
2. Devices such as routers and switches that forward packets are called **packet** _____.
3. In networking, a **protocol** defines the format, order, and actions taken for exchanging _____.
4. The computers running network applications at the edge of the Internet are called _____ **systems**.
5. Communication links in networks may use physical media such as fiber, copper, radio, or _____.
6. The **network edge** consists mainly of hosts such as clients and _____.
7. The part of the network composed of interconnected routers responsible for forwarding packets is called the **network** _____.
8. In a **client-server architecture**, the server is typically an _____ host with a permanent IP address.
9. In **peer-to-peer architecture**, peers both request services and _____ services to other peers.
10. A **process** communicating over the network is identified using an IP address together with a _____ number.
11. In socket communication, the socket acts as the **door between the application and the** _____ **layer**.
12. HTTP is an example of an _____ **layer protocol** used by the Web.
13. Email transfer across mail servers typically uses the protocol called _____.

14. The transport layer provides logical communication between application _____ running on different hosts.
15. The Internet transport protocol that provides reliable, ordered delivery of data is _____.
16. The Internet transport protocol that provides a lightweight, connectionless service is _____.
17. When sending data, the transport layer divides application messages into smaller units called _____.
18. The process of gathering data from multiple sockets at the sender is called _____.
19. The process of delivering received segments to the correct socket at the receiver is called _____.
20. The two main functions of the network layer are **routing** and _____.

Q2. Examples of vs questions

Compare the following paradigms against each other. Provide five main differences

- a. Circuit Switching vs Packet Switching**
- b. Forwarding vs Routing**
- c. Client–Server vs Peer-to-Peer architecture**
- d. Persistent HTTP vs Non-Persistent HTTP**
- e. Socket vs Port**
- f. Forwarding Table vs Routing Algorithm**

Q3. Problems

a. HTTP Response Time

A client wants to download a web page from a server. The page contains:

- **1 base HTML file**
- **4 referenced image objects**

Assume the following:

- **RTT between client and server = 80 ms**
 - **Transmission time for each object = 20 ms**
 - Ignore DNS lookup time.
- i. Calculate the total time required to download the entire page using **non-persistent HTTP**.
 - ii. Calculate the total time required using **persistent HTTP without pipelining**.

b. Multiplexing Example

A host is running **three applications** that send data simultaneously to the network.

Each application sends the following data:

- **Application A:** 600 bytes
- **Application B:** 1000 bytes
- **Application C:** 400 bytes

The transport layer protocol adds a **20-byte header** to every segment before passing it to the network layer.

Assume each application message is placed in **one segment**.

Questions

- a) How many **segments** are created by the transport layer?
- b) What is the **total amount of data transmitted** (including headers)?
- c) What percentage of the transmitted data is **header overhead**?

c. Packet Transmission Delay

A host wants to send a packet of **12,000 bits** over a link with transmission rate **3 Mbps**. Assume: **Propagation delay = 8 ms**, Ignore queuing and processing delays.

Questions:

- a) Calculate the transmission delay.
- b) Calculate the total end-to-end delay over this single link.

Q4. Subnet Allocation

Your company received the IP block **10.40.16.0/21** from the ISP.

To verify the assignment, the network administrator performed the following tests:

- **10.40.15.200 → Responded**
- **10.40.16.10 → Responded**

This raised questions about whether the assigned block is correct.

Your company must allocate subnets for the following departments:

Department	Required Hosts
Engineering	500
Marketing	220
HR	100
IT Support	40

- Use range calculations to determine whether 10.40.15.200 belongs to the block 10.40.16.0/21.
- If it does not belong to the block, determine which CIDR block most likely contains both IP addresses.
- Subnet the corrected block using VLSM to satisfy the host requirements of the four departments.

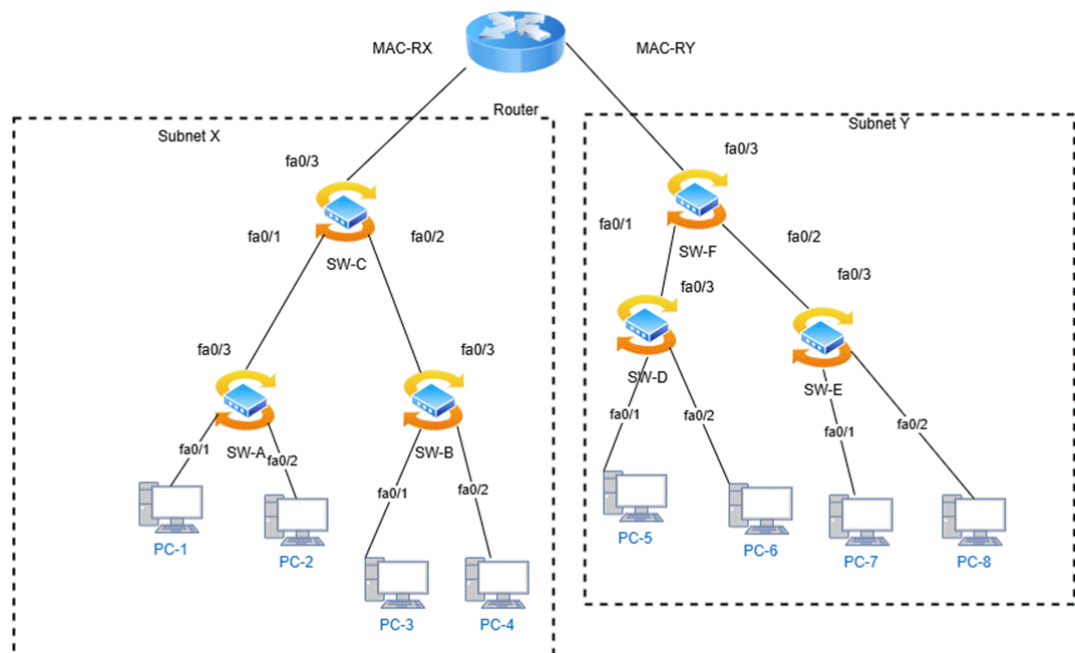
Fill in the table:

Department	Subnet Address	CIDR Mask
Engineering		
Marketing		
HR		
IT Support		

- d) After assigning these subnets, determine the largest single contiguous CIDR block that can still be allocated from the remaining address space.

Express the answer in CIDR notation.

Q5. ARP and Switch Tables



Consider the network topology shown in the figure.

Two subnets exist:

- **Subnet X:** PC-1, PC-2, PC-3, PC-4
- **Subnet Y:** PC-5, PC-6, PC-7, PC-8

The router connects the two subnets through interfaces **MAC-RX** and **MAC-RY**.

Initial conditions at **t = 0**:

- All **ARP tables (PCs and router)** are **fully up-to-date**
- All **switch MAC forwarding tables** are **empty**
- All PCs know the IP address of the router

- The router's routing table is complete
- **No DNS lookups are required**

a. At $t = 0.1$, PC-2 sends an IP segment to PC-5. At $t = 0.11$, the segment is successfully received by PC-5. What is the state of the MAC forwarding tables on all switches at $t = 0.2$?

Fill the following tables.

Switch	MAC Address	Port
SW-A		
SW-B		
SW-C		
SW-D		
SW-E		
SW-F		

b. At $t = 0.3$, PC-5 replies with an IP segment to PC-2. At $t = 0.31$, the segment is received by PC-2. What is the state of the MAC forwarding tables at $t = 0.4$?

Fill the following tables.

Switch	MAC Address	Port
SW-A		
SW-B		
SW-C		
SW-D		
SW-E		
SW-F		

- c. At $t = 0.5$, PC-1 sends an IP segment to PC-8. At $t = 0.51$, the segment arrives at PC-8. What is the state of the MAC forwarding tables at $t = 0.6$?

Fill the following tables.

Switch	MAC Address	Port
SW-A		
SW-B		
SW-C		
SW-D		
SW-E		
SW-F		